

— A / SKILLS

- LANGUAGES
- TypeScript · JS · Python · SQL · R
- AI ENGINEERING
- Anthropic API · Claude Code · agent orchestration · hybrid LLM + deterministic · PHI pipelines
- FRAMEWORKS
- SvelteKit 5 · React 19 · Vite · Bun · Vitest · R/Shiny · pandas · Chart.js · MapLibre
- PLATFORMS
- Cloudflare Workers · Pages · Durable Objects · D1 · KV · Access · Drizzle · Web Crypto · GitHub Actions
- DATA & FINANCE
- DCF + sensitivity modelling · derivatives pricing · interactive dashboards
- COMPLIANCE
- Privacy Act 1988 + APPs · My Health Records Act 2012 · MBS / HIA 1973

— B / EDUCATION

University of Queensland
B. Adv. Business (Hons), Finance
feb 2023 - dec 2027 (penultimate year)
BSAN2205 Machine Learning, BSAN3204 Data Viz, BSAN4204 Business Analytics, Corporate Finance, Asset Pricing, Derivatives.

— C / ADDITIONAL

- SOCIETIES
- UQ Economics Society + Business Analytics Network Association (2023-).
- LANGUAGES
- English (native) · Japanese (JLPT N4)
- VOLUNTEERING
- Community BBQ for people experiencing homelessness (2020-2022).

— 01 / SUMMARY

Founder of [Tek Monkeys](#), a Brisbane IT consultancy serving a paediatric orthopaedic practice, where I designed and deployed [Itemate](#), an MBS billing validator pairing the Anthropic Claude API with a deterministic matcher and a server-side PHI-free data flow. I build production TypeScript and LLM applications on Cloudflare. Third-year UQ Business (Honours) student, Business Analytics + Finance double major, graduating 2027.

— 02 / EXPERIENCE

Founder & AI Developer @ [Tek Monkeys](#) 2024 - present
brisbane it consultancy · paediatric orthopaedic practice · Brisbane, QLD

- Scoped, designed, and shipped [Itemate](#), the practice's MBS billing validator, after workflow interviews with both surgeons; delivered as a managed service with ongoing iteration.
- Sole technical contact for the practice: IT support, hardware, and software, managed within the Privacy Act 1988 (APPs) and the My Health Records Act 2012.

Hospitality & Customer Service @ [Brisbane Airport](#) 2023 - feb 2026
three years across four BNE food & beverage venues spanning the domestic and international terminals

- Three years of high-volume customer service across four food & beverage venues spanning the domestic and international terminals, alongside full-time study.
- Progressed from counter service to specialty coffee at an award-winning roaster's flagship.

— 03 / SELECTED PROJECTS

Monkey Barrel [real-time multiplayer web arcade](#) apr 2026 - present
[arcade.nickwfraser.dev](#)

TypeScript SvelteKit 5 Cloudflare Workers Durable Objects D1 Web Crypto Bun

- Built and deployed a multiplayer party-game platform hosting 12+ games plus a 6,400-line idle-RPG sub-app (9 zones, 180 stages). Coordinated authoritative game state through Durable Objects on the WebSocket Hibernation API for near-zero idle cost; ~1,000 games played since launch.
- Architected an actor-model room hierarchy (one Durable Object per room) with PBKDF2 + D1-backed session auth via Web Crypto; diagnosed and hotfixed a production incident where Workers' 100k PBKDF2 iteration cap blocked NIST G1 compliance.
- Used multi-agent AI code audits (Claude Code) to catch a Durable Object hibernation regression across 5 room subclasses; shipped the fix as a unified connection model with 18 passing integration tests.

Itemate [MBS billing validator](#) 2026 - present
[itemate.nickwfraser.dev](#)

TypeScript SvelteKit 5 CF Pages + D1 Drizzle Anthropic Claude API Vitest

- Built and deployed Itemate for a paediatric orthopaedic practice (two surgeons), pairing the Anthropic Claude API (Haiku for dev, Sonnet for prod) for clinical note normalisation with a deterministic keyword-overlap retrieval pass over a 6,039-item MBS index. Deployed at the edge behind Cloudflare Access (Google OAuth).
- Engineered a regex-anchored de-identification pipeline that strips PHI server-side before any LLM call; shipped 97 unit tests with GitHub Actions CI. No patient data persists on the server.

gocard.nickwfraser.dev

- TypeScript
- SvelteKit 5
- CF Pages + Workers
- D1 + KV
- Chart.js
- MapLibre GL

- Built and shipped the BSAN4204 Business Analytics research project, combining Go Card trip-history analysis with a live mode-choice calculator comparing public transport against driving costs after Translink's 50¢ flat-fare reform.
- Engineered a daily fuel-data pipeline as a sibling Cloudflare Worker cron against the QLD FPQ API with KV caching and D1 history retention; validated a pure mode-choice engine across 24 unit tests to within \$0.50 per week of manual spreadsheet calculations.